Human fall detection using accelerometer and gyroscope sensors in unconstrained smartphone positions

Abstract

This study explored several methods for detecting body falls based on the data captured by the sensors (accelerometer and gyroscope) built in a smartphone carried by a person. The data for this study were collected by recording many sample units from each of the following human activity categories: stand-fall, walk-fall, stand-jump, stand-sit, stand, and walk. Several time-series data captured by the sensors were used as human motion features. One of the challenges of this study was the existence of human body motions whose features resembled those of body falls. In addition, unfixed smartphone positioning made human body falls harder to detect and can lead to high rate of misclassification (not detected as fall). This incident can caused serious bone fracture or even death if the person not handled as immediately as possible because of misclassification. To address this problem, we modified Resultant Acceleration and ∠ Y formulas to address the problem of unconstrained smartphone positions. We proposed to combine five methods such as AGVeSR, Alim, ∠α, GyroReDi, and AGPeak to build a robust detector model to reduce the misclassification. Human fall detection using smartphone sensors is an important research area that aims to improve personal safety by identifying falls and enabling timely assistance. This study investigates several methods for detecting human falls using data collected from the accelerometer and gyroscope sensors embedded in smartphones carried by individuals. The primary objective is to develop a reliable fall-detection system that can recognize fall events under different smartphone positions and everyday movement conditions.

The data used in the study were collected from multiple human activity categories, including stand-fall, walk-fall, stand-jump, stand-sit, standing, and walking. These activities were selected to evaluate the ability of the proposed system to distinguish actual falls from normal human movements. Time-series measurements obtained from the accelerometer and gyroscope were used as features to characterize body movements and identify patterns associated with falls.

One of the major challenges addressed in this research is the similarity between certain normal activities and actual falls. Movements such as jumping or sitting down quickly can produce sensor patterns resembling those generated during a fall, potentially leading to incorrect classifications. Another significant challenge is the unconstrained positioning of smartphones. Since users may carry their phones in different orientations and locations, the sensor readings can vary considerably, making reliable fall detection more difficult. Failure to recognize a genuine fall may delay assistance and increase the risk of serious injuries.

To address these challenges, the researchers modified the Resultant Acceleration and ∠Y formulas to improve detection when smartphone positions are not fixed. Furthermore, they proposed a combined detection model incorporating five methods: AGVeSR, Alim, ∠α, GyroReDi, and AGPeak. These methods were integrated using a decision-fusion approach to improve the overall reliability of the system and reduce the likelihood of misclassifying fall events.

The experimental results demonstrated that combining accelerometer and gyroscope data achieved an accuracy improvement of more than 15% compared with using the accelerometer alone. Additionally, the decision-fusion technique, which combined the outputs of the five methods through voting, improved the accuracy rate by up to 4.15%. These findings highlight the advantages of combining multiple sensor types and detection methods when designing systems that must operate under varying real-world conditions.

Overall, the study demonstrates the potential of smartphone-based motion sensing for developing more reliable human activity and fall-detection systems. The findings emphasize the importance of handling variations in smartphone orientation, distinguishing falls from similar everyday activities, and combining information from multiple sensors. Although the research focuses on human fall detection rather than urban climate hazards, its approach to analysing accelerometer and groscope time-series data may offer useful methodological insights for the **ClimatePath ideathon project**, particularly when investigating pedestrian movement patterns. However, additional research would be required to determine whether changes in pedestrian movement can reliably indicate environmental hazards such as street-level waterlogging.

Author keywords

Accelerometer; Gyroscope; Human fall detection; Unconstrained smartphone positions